

(2021~2022 学年第 2 学期)

适用专业年级: 软件工程 2020 级 学号: 姓名:

四川大学各级各类考试的监考人员，必须严格执行《四川大学考试工作管理办法》、《四川大学考场规则》和《四川大学监考人员职责》。有违反学校有关规定的，严格按照《四川大学教学事故认定及处理办法》进行处理。

题号	1	2	3	4	5				卷面 成绩
得分									
阅卷时间									

2. 考试结束, 请将答题纸和试卷一并交给监考老师。



评阅教师	本题得分

1. 目前在数据库领域采用最广泛的数据模型是 ()
A. 层次模型 B. 网状模型 C. 关系模型 D. 面向对象模型
2. 数据库中产生数据不一致的根本原因是 ()。
A. 数据存储空间太大 B. 没有严格保护数据 C. 未对数据进行完整性控制 D. 数据冗余
3. 我们在关系数据库中定义外码主要用于保证数据的 ()
A. 实体完整性 B. 参照完整性 C. 用户定义的完整性 D. 语义完整性
4. 事务中的所有操作要么都生效, 要么都不生效, 是事务的 ()
A. 原子性 B. 一致性 C. 隔离性 D. 持续性
5. 下列 SQL 子句真实执行顺序正确的是 ()

本题共 4 页，本页为第 1 页
教务处试题编号：

- A. SELECT-FROM-GROUP-WHERE-ORDER
B. FROM-SELECT-WHERE-GROUP-ORDER
C. FROM-WHERE-GROUP-SELECT-ORDER
D. SELECT-FROM-WHERE-GROUP-ORDER
6. 设关系 R(A,B,C)和 S(B,C,D)，下列关系代数表达式不成立的是 ()
A. $\pi_A(R) \bowtie \pi_D(S)$ B. $R \cup S$ C. $\pi_B(R) \cap \pi_B(S)$ D. $R \bowtie S$
7. ER 图设计是在数据库设计的 () 阶段进行的。
A. 需求采集 B. 概念设计 C. 逻辑设计 D. 物理设计
8. 如果一个关系模式的所有候选码都只有一个属性，则此关系模式至少达到 ()
A. 2NF B. 3NF C. BCNF D. 4NF
9. 对关系模式的无损分解能够确保: ()
A. 分解后关系达到 3NF B. 保持函数依赖 C. 分解后关系的自然连接和原关系相等 D. 没有任何信息损失
10. 下列哪一个操作可以收回对表的授权? ()
A. DELETE D. DROP B. REMOVE C. REVOKE

评阅教师	本题得分

二. 查询. (4分×10=40分)

有如下数据库模式 (下划线标记的是主码) .

user (ID, name, team, phone) //用户 (ID, 姓名, 小组, 电话)

dev (dNo, dName, type) //设备 (设备编号, 名称, 类型)

adm (NO, name, salary, store) //库房管理员 (编号, 姓名, 薪水, 库房)

borrow (ID, dNo, date, NO) //设备借用 (用户 ID, 设备编号, 日期, 经手库管编号)

1、请使用关系代数完成以下查询:

- 列出小组为“锦江”的用户的 ID 和姓名
- 列出名为“李四”的用户借用过的设备的编号和名称
- 列出 ID 为 777 的用户借用过且经手库管编号为 999 的设备的编号
- 列出满足如下条件的用户的 ID 和姓名: 此用户借用了 ID 为 888 的用户借用过的所有设备

2、通过 SQL 完成如下查询:

- 列出薪水高于 5000 的管理员姓名.
- 列出管理员平均薪水高于 5000 的库房
- 列出在各自所属库房中薪水最高的管理员的姓名.
- 列出在'2022-6-1'和'2022-6-30'之间被 ID 为 666 的用户借用过的设备编号和名称

注: 试题字迹务必清晰, 书写工整。

本题共 4 页, 本页为第 2 页
教务处试题编号:

- (5) 列出从来没有由编号为 555 的库管经手借用过设备的用户的姓名
 (6) 列出满足如下条件的用户 ID 和姓名: 该用户借用过所有曾经被 555 号管理员经手借出过的设备。

评阅教师	本题得分

三. 规范化 (7分×3=21分)

请为对下列每个关系模式完成:

- 求出所有的候选码
- 判定它的范式级别并说明理由.
- 如果它的范式级别未达到 3NF, 请给出一个保持函数依赖的无损分解使其满足第三范式.

- 运动员 (ID, 姓名, 生日, 团队, 领队电话)
 一个运动员只能加入一个团队; 一个团队只有一个领队, 但一个人可以担任多个团队的领队. 领队只有一个电话.
- Shop (NO, name, addr, owner, item)
 一个商店(shop)有一个店主(owner), 一个地址(addr), 多个商品(item)
- R (U, F), U={A, B, C, D}, F={BC, (A,B)D, CB}

评阅教师	本题得分

四. 并发控制与恢复 (9分)

有如下关于事务 T1 和 T2 的一个调度.

step	T1	T2
1	Read(C)	
2	C = C - 5	
3		Read(B)
4		B = B - 10
5	Write(C)	
6		Write(B)
7		Read(A)
8		A = A - 2
9		Write(A)
10	Read(A)	
11		Commit
12	A = A / 2	
13	Write(A)	
14	Commit	

- 调度是冲突可串行化的吗? 如果是, 它等效于 T1, T2 还是 T2, T1 的串行顺序? 如果不是, 给出简要理由. (5分)
- 假定在步骤 1 时, A, B, C 的值分别是 100, 100, 100, 请按“<Ti-Start>, <Ti, 数据名, 旧值, 新值>, <Ti-Commit>”

注: 试题字迹务必清晰, 书写工整。

本题共 4 页, 本页为第 3 页
 教务处试题编号:

课程名称： 任课教师： 学号： 姓名：

这样的格式为这个调度写出日志(不记录读操作), 如 “<T1-Start>,<T2,B,45,50>……” (2 分)

3. 如果在步骤 12 后 13 前发生了系统崩溃, 请给出系统重启数据库完成恢复后 A,B,C 的值. (2 分)

评阅教师	本题得分

五. 数据库设计 (20 分)

为一个医院的住院部建立数据库管理系统, 管理如下数据:

- (1) Patient (病人) 包括: ID, name(姓名), age(年龄), gender(性别)
- (2) Nurse (护士) 包括: nurse_no(护士编号), name(姓名)
- (3) Doctor (医生) 包括: doctor_no(医生编号), name(姓名)
- (4) Ward (病房) 包括: ward_no(房号), capacity(容量)

一个病房可容纳多个病人; 一个病房可被多个护士管理, 一个护士可管理多个病房; 一个医生可担任多个病员服的主治医生, 但一个病人智能有一个主治医生。

1. 请给出 ER 图设计 (8 分)
2. 根据 ER 图给出对应的满足 3NF 的关系模式设计 (通过下划线标记主码) (8 分)
3. 使用 SQL 语句查找被护士 “艾美丽” 管理的病人姓名和年龄 (4 分)

ER图设计
模式
查询

(2018-2019 学年第 2 学期)

适用专业年级: 软件工程 2017 级 学号: _____ 姓名: _____

本题共 6 页，本页为第 1 页
教务处试题编号：

6. When mapping from an ER model to a relational model, a strong entity is mapped into a (B)
 (A) column (B) table (C) key (D) row
7. What attributes does a subclass have? (D)
 (A) Just the attributes from the superclass
 (B) A subset of the attributes of its superclass
 (C) None of the attributes of its superclass
 (D) All the attributes of its superclass, and possibly more
8. Let $R(A,B,C,D)$ be a relation schema, and $A \rightarrow B, C \rightarrow D$, then (D) is wrong.
 (A) $AC \rightarrow B$ (B) $AC \rightarrow D$ (C) $AD \rightarrow B$ (D) $AD \rightarrow BC$
9. In a database system, which part is it to provide data consistency? (C)
 (A) the DBA (B) then user (C) the DBMS (D) the application program
10. in 2PL protocol, at (B) stage, A transaction may not obtain locks, but can release locks.
 (A) Growing phase (B) Shrinking phase (C) Committed (D) Aborted

评阅教师	本题得分

II. Queries. (5 points each; 40 total)

Consider the relational database of a banking enterprise with the following relation schemas, where the primary keys are underlined.

branch (branch_name, branch_city, assets)
 customer (customer_name, customer_street, customer_city)
 loan (loan_number, branch_name, amount)
 borrower (customer_name, loan_number)
 account (account_number, branch_name, balance)
 depositor (customer_name, account_number)

1、Give a relational algebra expression for each of the following queries:

(1) Find all loan number for loans made at the 'Perryridge' branch (that is the name of a certain branch) with loan amounts greater than \$1000

$\Pi_{\text{loan_number}} \sigma_{\text{branch_name}='Perryridge' \text{ and } \text{amount} > 1000} (\text{loan})$ ✓

(2) Find names of all customers who have accounts in **all** branches located in 'Brooklyn'.

~~Find the max account balance in branch located in 'Brooklyn'~~

(3) Find the max account balance in branch located in 'Brooklyn'

$\text{branch_name} \bowtie \mathcal{G}_{\text{max}(\text{balance})} (\sigma_{\text{branch_city}='Brooklyn'} (\text{account} \bowtie \text{Branch}))$

2、Write SQL statements to perform the following commands:

(1) Find the name of all customers who have accounts but not loans.

(Select customer_name from depositor) except (Select customer_name from borrower)

课程名称:

任课教师:

学号:

姓名:

(2) Find the name of all customers who have accounts in all branches located in 'Brooklyn'.

(3) Find the name of all branches that have assets greater than those of at least one branch located in 'Brooklyn'.

```
Select branch_name from branch where assets>some(select assets from branch where branch_city='Brooklyn')
```

(4) Find the names of all branches where the average account balance is more than \$2000.

```
Select branch_name from account group by branch_name having avg(balance)>2000
```

(5) Find all customers that live in the same city with branches they open accounts.

```
Select customer_name, customer_city from ((customer natural join depositor) natural join account) natural join branch where customer_city= branch_city
```

评阅教师	本题得分

III. Normalization (20 points total)

1. Consider the following relational schema:

```
student_graduation_project=(sID, name, iID, iName, iTitle, year, projectName, grade)
```

It contains information about graduation designs of students. The Attribute iID is the ID of Instructor, The Attribute iName is the name of Instructor, The Attribute iTitle is the title of Instructor, We have the following information on the attributes:

- Each student has only one instructor as graduation design advisor.
- Each student has only one graduation design project.
- Each instructor can advisor many students in graduation design.
- Students can take part in same projects.

The following is an instance of the schema:

sID	Name	iID	iName	iTitle	projectName	grade
11	Cheng	101	Yuan	Professor	Face Image Recognition	B
12	Zhang	101	Yuan	Professor	Machine Learning	A
13	Liu	102	Cheng	Instructor	Face Image Recognition	A
14	Yang	102	Cheng	Instructor	Machine Learning	C
15	Li	103	Zhou	Professor	Speech Recognition	D

(1) Based on above, indicate for each of the following potential functional dependencies, whether it is indeed an FD or not. (3 points)

A. sID → Name B. sID → iID C. iID → grade

D. iName → iID E. ID → (projectName, grade) F. projectName → grade

True: A, B, E 选对一个给 1 分

(2) Based on above, indicate for each of following sets of attributes whether it is a candidate key for student_graduation_project or not. (3 points)

A. {sID} B. {sID, iID} C. {sID, projectName}

D. {name} E. {name, iName} F. {iID, projectName}

Candidate Key: A

(3) Is the relation schema student_graduation_project in BCNF? Why? Is it in 3NF? Why? If it is not in 3NF, bring it to a set of relations at least in 3NF; specify primary keys and referential integrity constraints for each relation. (4 points)

注: 试题字迹务必清晰, 书写工整。

本题共 6 页, 本页为第 3 页
教务处试题编号:

Not in BCNF and not in 3NF. {sID} is the candidate key of it, iID $\rightarrow$ iName(其他的也行), iID is not a super key and title is not in any candidate key.

This gives the following 3NF relations: (命名可以不同)

instructor(iID, iName, iTitle)

student_project(sID, name, iID, projectName, grade) (iID) $\rightarrow$ instructor

2. Consider a relation R(A, B, C, D, E) with the set of Functional Dependencies

$F = \{A \rightarrow BD, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$

(1) Give all candidate keys of R. [3 points]

AC, BC, DC

(2) Give a canonical cover of F. [3 points] (ANSWER IN NOT UNIQUE)

$F_c = \{A \rightarrow B, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$

(3) Is R in 3NF? explain why if it is or decompose it into 3NF if not. [4 points]

(ANSWER IN NOT UNIQUE)

No, decompose it into $\{R_1(AB), R_2(BCE), R_3(BD), R_4(DA)\}$

评阅教师	本题得分

IV. Concurrent Control (10 points total)

Consider two transactions T1 and T2.

T1: read(B); A=B+1; write(A)

T2: read(A); B=A+1; write(B)

Consider the concurrent schedule of transactions T1 and T2:

T1	T2
read (B)	
A=B+1	
	read (A)
write(A)	
	B=A+1
	write (B)
Commit	
	commit

1. Give the result of above schedule. (initial value A=2, B=2). (3points)

2. Is the schedule correct or not? Why? (3points)

3. Add lock and unlock instructions satisfying 2PL to Transactions T1 and T2, to implement a serializable schedule. (4points)

Answer:

1. A=3, B=3

2. 不正确。多个事务的并发执行是正确的，当且仅当其结果与按某一次序串行地执行时的结果相同，称这种调度策略为可串行化的调度。T1、T2 串行调度结果为：1. A=4, B=3; 2. A=3, B=4

3. T1 T2

SLOCK B

read(B)

A=B+1

注：试题字迹务必清晰，书写工整。

本题共 6 页，本页为第 4 页
教务处试题编号：

```

XLOCK A
SLOCK A
wait
write(A)
wait
UNLOCK B
wait
UNLOCK A
wait
read(A)
XLOCK B
B=A+1
write(B)
UNLOCK A
UNLOCK B
    
```

评阅教师	本题得分

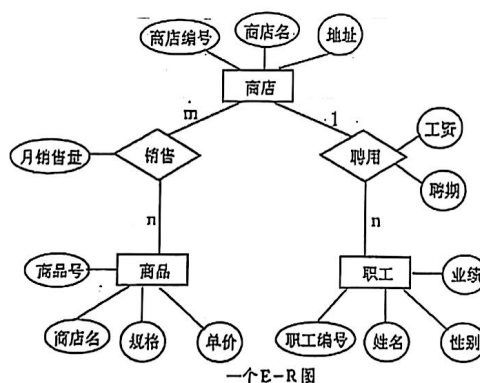
V. Database Design (20 points total)

Entity Store is associated with store number, store name and address etc. Entity commodity(商品) is associated with commodity number, commodity name, specification and price. Entity employee is associated with employee no, employee name, sex and performance. Each commodity can be sold in many stores and each store may sold many kinds of commodity. For each commodity in a certain store, the sale quantity must be recorded per month. For each store, there are many employees. For each employee, he can works for only one store. For each employee who works in a certain store, salary and work period must be recorded.

1. Construct an E-R diagram that captures the information above. (10 points)
2. Convert the E-R diagram to 3NF relations. Specify keys and referential integrity constraints. (10 points)

Answer:

1.



2.

Store(Store_number, store_name, address)

Employee(employee_no, employee_name, sex, performance)

commodity (commodity_number, commodity_name, specification, Price)

sale (Store_number, commodity_number, MonthQuantity)

Store_number is a foreign key referring store(store_number).

注：试题字迹务必清晰，书写工整。

本题共 6 页，本页为第 5 页
教务处试题编号：

课程名称:

任课教师:

学号:

姓名:

commodity_number is a foreign key referring commodity(commodity_number).

Hire(employee_no,Store_number, During, salary)

Store_number is a foreign key referring store(store_number).

employee_no is a foreign key referring employee(employee_no).

四川大学期末考试试题（闭卷）

（2017~2018 学年第 2 学期）

课程号：311038040 课程名称：数据库系统（A 卷） 任课教师：刘一静、吴成恩、屈立筌、张天庆

适用专业年级：软件工程 2016 级 学号： 姓名：

考试须知									
四川大学学生参加由学校组织或由学校承办的各级各类考试，必须严格执行《四川大学考试工作管理办法》和《四川大学考场规则》。有考试违纪作弊行为的，一律按照《四川大学学生考试违纪作弊处罚条例》进行处理。 四川大学各级各类考试的监考人员，必须严格执行《四川大学考试工作管理办法》、《四川大学考场规则》和《四川大学监考人员职责》。有违反学校有关规定的，严格按照《四川大学教学事故认定及处理办法》进行处理。									
题 号	1	2	3	4	5				卷面成绩
得 分									
阅卷时间									

注意事项： 1. 请务必将本人所在学院、姓名、学号、任课教师姓名等信息准确填写在试题纸和添卷纸上；

2. 考试结束，请将试题纸、添卷纸和草稿纸一并交给监考老师。



评阅教师	本题得分

I. Simple choice. (2 points each; 10 total)

1	2	3	4	5

- What does ACID stand for in the context of DBMS transactions? A
 - Atomicity, Consistency, Isolation, and Durability
 - Analysis Console for Intrusion Databases
 - Atomicity, Consistency, Isolation, and Data
 - Automatic Classification and Interpretation of Data

2. A candidate key is: C

- a set of one or more attributes that allow us to identify uniquely a tuple in the relation
- functionally dependent on non-key attributes
- the minimal super key (no proper subset of the super key is a super key)

注：试题字迹务必清晰，书写工整。

本题共 6 页，本页为第 1 页
教务处试题编号：

(d) the primary key of a relation

3. Which pair of SQL keywords is used to create and delete a table? A

- (a) CREATE, DROP
- (b) CREATE, DELETE
- (c) CREATE, ALTER
- (d) INSERT, DROP

4. Which pair of SQL keywords is used to create and delete records in a table? D

- (a) CREATE, DROP
- (b) CREATE, DELETE
- (c) CREATE, ALTER
- (d) INSERT, DELETE

5. A lossless join decomposition of a relation means: C

- (a) none of the attributes are lost
- (b) no functional dependencies are lost
- (c) the natural join of the relations in the decomposition produces the original relation
- (d) no information of any kind is lost

评阅教师	本题得分

II. Queries. (5 points each; 40 total)

An employee training database contains following relational schemas (primary keys are underlined):

employee(ID, name, address, salary)

course(course_id, title, hours)

section(course_id, sec_id, start_date, finish_date)

takes(employee_id, course_id, sec_id, grade)

1. Give a relational algebra expression for each of the following queries: 注意可能有多种

解法

(1) List the IDs and names of all employees who have taken a course titled "Company Culture".

$\Pi_{employee.ID, name} \sigma_{employee.ID = takes.employee_ID \wedge takes.course_id = course.course_id \wedge title = 'Company Culture'} (employee \times takes \times section)$

(2) List the IDs and names of all employees who have taken all courses that the employee with ID "dev101" has taken.

$\Pi_{employee.ID, name} \sigma_{employee.ID = takes.employee_id} (\Pi_{employee_id, course_id} (takes) \div \Pi_{course_id} \sigma_{employee_id = 'dev101'} (takes) \times employee)$

(3) List the IDs of all employees who have taken more than five courses.

$\Pi_{employee_id} \sigma_{number_course > 5} (employee_id \bowtie count_distinct(course_id) as num_course (takes))$

2. Write SQL statements to perform the following commands:

注：试题字迹务必清晰，书写工整。

本题共 6 页，本页为第 2 页
教务处试题编号：

- (1) List the IDs and names of all employees who have NEVER taken the course titled "Business Decision".

```
select id, name
  from employee
 where ID not in
 (select employee_id
   from takes, course
   where takes.course_id = course.course_id and title=' Business Decision');
```

- (2) List the IDs and titles of all courses whose title begin with "Business".

```
select course_id, title
  from course
 where title like 'Business%';
```

- (3) List the ID and name of the employee who have taken the course titled "Business Decision" and get the highest grade.

```
select id, name
  from employee e, takes t, course c
 where ID=t.employee_id and t.course_id = c.course_id title=' Business Decision' and grade =
 (select max(grade)
  from takes t, course c
  where t.course_id=c.course_id and title=' Business Decision');
```

- (4) List the IDs and names of all employees who have taken all courses that the employee with id "dev101" has taken.

```
select ID, name
  from employee e
 where not exists
 ((select course_id
   from takes
   where employee_id='dev101')
 except
 (select course_id
  from takes
  where e.ID=employee_id));
```

- (5) List the employee names and the average grades of courses they have taken.

```
select name, avg(grade)
  from employee, takes
 where ID=employee_id
 group by ID, name;
```

评阅教师	本题得分

III. Normalization (20 points total)

1. Consider the following relational schema:

Articles=(ID, title, journal, issue, year, start page, end page, TR_ID)

注: 试题字迹务必清晰, 书写工整。

本题共 6 页, 本页为第 3 页
教务处试题编号:

It contains information on articles published in scientific journals. Each article has a unique ID, a title, and information on where to find it (name of journal, what issue, and on which page). Also, if results of an article previously appeared in a "technical report" (TR), the ID of the technical report can be specified. We have the following information on the attributes:

- For each journal, an issue with a given number is published in a single year.
- The end page of an article is determined by the start page.
- There is never more than one article on a single page.

The following is an instance of the schema:

ID	title	journal	issue	year	start_page	end_page	TR_ID
2	Cuckoo Hashing	JAlg	51	2015	68	80	87
17	Time Serial DM	JAlg	48	2014	69	85	62
17	Time Serial DM	JAlg	48	2014	69	85	78
23	P vs. NP resolved	SICOMP	23	2016	3	7	98
25	P vs. NP resolved	JACM	46	2016	1	5	98

- (1) Based on above, indicate for each of the following potential functional dependencies, whether it is indeed an FD or not. (3 points)

A. $ID \rightarrow title$ B. $start_page \rightarrow end_page$ C. $(journal, issue) \rightarrow year$
D. $title \rightarrow ID$ E. $ID \rightarrow (start_page, end_page, journal, issue)$ F. $TR_ID \rightarrow ID$
True: A, C, E 选对一个给 1 分

- (2) Based on above, indicate for each of following sets of attributes whether it is a candidate key for **Articles** or not. (3 points)

A. {ID} B. {ID, TR_ID} C. {ID, title, TR_ID}
D. {title} E. {title, year} F. {start_page, journal, issue}
Candidate Key: B

- (3) Is the relation schema **Article** in **BCNF**? Why? Is it in **3NF**? Why? If it is not in 3NF, bring it to a set of relations at least in 3NF; specify primary keys and referential integrity constraints for each relation. (5 points)

Not in BCNF and not in 3NF. {ID, TR_ID} is the candidate key of it, $ID \rightarrow title$ (其他的也行), ID is not a super key and title is not in any candidate key.

This gives the following 3NF relations: (命名可以不同)

Journals(journal, issue, year)

Articles(ID, title, journal, issue, start_page, end_page)

FK: (journal, issue) → Journals

Attive_TRs(ID, TR_ID) FK: ID → Articles

2. Consider the relation schema $R = (A, B, C, D, E, F)$ and the set of functional dependencies $F = \{A \rightarrow B, A \rightarrow C, BC \rightarrow E, BC \rightarrow D, E \rightarrow F, BC \rightarrow F\}$

- (1) List the candidate key(s) for R. Write 'none' if you think there are no candidate keys. (3 points)

Candidate key: (A)

- (2) List the FDs in F that violates BCNF. (3 points)

$BC \rightarrow E, BC \rightarrow D, E \rightarrow F, BC \rightarrow F$

- (3) Is R in 3NF? Why? (3 points)

$BC \rightarrow E, BC \rightarrow D, E \rightarrow F, BC \rightarrow F$ (写一个就行): LHS is not a super key, and RHS is not in any candidate key.

评阅教师	本题得分

IV. Concurrent Control (10 points total)

Consider the concurrent schedule of transactions T1 and T2.

T1	T2
read (A)	
	read (B)
	read (A)
	write (A)
	write (B)
write (A)	
	commit
commit	

1. Is the schedule conflict serializable? If so, give an equivalent serial schedule. If not, give an explain briefly. (5 points)

No. it is not conflict serializable. 3分 T1's read(A) and write(A) is conflict instruction to T2's write(A). 2分

2. Add lock and unlock instructions to Transactions T1 and T2, so that they observe the two-phase-locking protocol. Can it results in a deadlock? (5 points)

T1: lock-X(A), read(A), write(A), unlock(A) 2分

T2: lock-X(B), lock-X(A), read(B), read(A), write(A), write(B), unlock(A), unlock(B). 2分

It cannot results in a deadlock, for T1 just lock only one item. 1分

评阅教师	本题得分

V. Database Design (20 points total)

To build a database to handle information about law firms(律师事务所). Each law firm is identified by a name that is unique within a city. A lawyer is identified by an ID, and have a name, address, and salary. A lawyer may be employed by at most one firm. A customer is also identified by an ID, and have a name, address and phone number. A firm can serve a number of customers, and a customer can have a number of law firm serves. Each lawyer handles a load of cases, which involves different customers, identified by a case number. Cases may be handled by more than one lawyer, even if the lawyers work for different firms.

- Construct an E-R diagram that captures the information above. (10 points)
- Convert the E-R diagram to 3NF relations. Specify keys and referential integrity constraints. (10 points)

评分要点：1. 实体集 5 分，联系集及约束 5 分；2. 关系模式转换 5 分，主码约束 3 分，外码 2 分。

entity sets:

law_firm(name, city)

lawyer(ID, name, address, salary)

注：试题字迹务必清晰，书写工整。

本题共 6 页，本页为第 5 页
教务处试题编号：

课程名称: _____ 任课教师: _____ 学号: _____ 姓名: _____

customer(ID, name, address, phone)

case(case_number, description) (除 case_number 外,其他属性可选)

relationship sets:

employ lawyer: law_firm N:1

serve customer:law_firm N:M

handle lawyer:case N:M

involve customer:case N:M

关系模式:

law_firm(name, city)

lawyer(ID, name, address, salary, firm_name, firm_city) FK: (firm_name, firm_city) -> law_firm

customer(ID, name, address, phone)

case(case_number, description)

serve (customer ID, firm_name, firm_city) FK1:customer_ID ->customer (ID) FK2:

(firm_name, firm_city) -> law_firm

handle(lawyer_ID, case_number) FK1:lawyer_ID ->lawyer(ID) FK2:case_number -> case

involve (customer_ID, case_number) FK1:customer_ID ->customer(ID) FK2:case_number
-> case

四川大学期末考试试题（闭卷）

（2017~2018 学年第 2 学期）

B 卷

课程号: 311038040 课程名称: 数据库系统 任课教师: _____

适用专业年级: 软件工程 2016 级 学号: _____ 姓名: _____

考生承诺

我已认真阅读并知晓《四川大学考场规则》和《四川大学本科学生考试违纪作弊处分规定（修订）》，郑重承诺：

- 1、已按要求将考试禁止携带的文具用品或与考试有关的物品放置在指定地点；
- 2、不带手机进入考场；
- 3、考试期间遵守以上两项规定，若有违规行为，同意按照有关条款接受处理。

考生签名: _____

题 号	一(10%)	二(40%)	三(20%)	四(10%)	五(20%)
得 分					
卷面总分			阅卷时间		

注意事项: 1. 请务必将本人所在学院、姓名、学号、任课教师姓名等信息准确填写在试题纸和添卷纸上；

2. 请将答案全部填写在本试题纸上；

3. 考试结束，请将试题纸、添卷纸和草稿纸一并交给监考老师。

评阅教师	得分

I. Simple choice. (2 points each; 10 total)

1	2	3	4	5

1. Which built-in SQL function computes the number of rows in a table?

- (a) AVG
- (b) MAX
- (c) COUNT
- (d) SUM

2. Which one of the following is **not** true?

- (a) Every relation schema that's in BCNF is also in 3NF.
- (b) Every relation schema that's in 3NF is also in 1NF.
- (c) A relation schema R is in 1NF if the domains of all attributes of R are atomic.
- (d) Decomposition of a 3NF relation into BCNF always preserves functional dependencies.

3. $F = \{A \rightarrow B, B \rightarrow C\}$, which functional dependencies **cannot** be implied?

- (a) $A \rightarrow C$
- (b) $A \rightarrow BC$
- (c) $B \rightarrow BC$
- (d) $C \rightarrow B$

4. Which of the following is not a property of transaction?

- (a) Atomicity
- (b) Serializability
- (c) Durability
- (d) Isolation

5. The relation $r_1(R)$ has 10 tuples, and $r_2(R)$ has 8 tuples, the number of tuples of $r_1 \cup r_2$ can be:

- (a) 2
- (b) 8
- (c) 15
- (d) 80

评阅教师	得分

II. Queries. (5 points each; 40 total)

The following relational schemas store information about students and research projects that they have taken part in. (The primary keys are underlined):

student (ID, name, dept_name)

project(project_no, title)

s_p(student_id, project_no, grade)

1. Give a relational algebra expression for each of the following queries:

- (1) List the IDs and names of all students who have taken part in the project titled "Wechat Programming".
- (2) List the IDs and names of all students who have taken all projects that the student with ID "CS101" has taken part in.
- (3) List the IDs of all students who have taken part in more than five projects.

2. Write SQL statements to perform the following commands:

- (1) List the IDs and names of all students who have **NEVER** taken part in the project titled "Wechat Programming".
- (2) List the project numbers and titles of all projects whose title begin with "Program".
- (3) List the ID and name of the employee who have taken part in the project titled "Programming of Wechat" and get the highest grade.
- (4) List the IDs and names of all students who have taken part in all projects that the student with ID "CS101" has taken part in.
- (4) List the students' names and the average grades of projects they have taken part in.

评阅教师	得分

III. Normalization (20 points total)

1. Consider the following relational schema:

student_dept=(ID, name, tot_credit, dept_name, building, budget)

It contains information about students and their departments in a university. A student studies in just one department, and a department has exactly a building and budget. The following is an instance of the schema:

ID	name	tot_credit	dept_name	building	budget
00128	Zhang	102	Comp. Sci.	Taylor	100000
12345	Shankar	32	Comp. Sci.	Taylor	100000
19991	Brandt	80	History	Painter	50000
23121	Chavez	110	Finance	Painter	120000
44553	Peltier	56	Physics	Watson	70000
45678	Levy	46	Physics	Watson	70000

- (1) Identify **functional dependencies** of *student_dept* based on above. (3 points)
 - (2) Identify the **candidate key(s)** of *student_dept*. (3 points)
 - (3) IS the relation schema *student_dept* in **BCNF**? Why? Is it in **3NF**? Why? If it is not in 3NF, bring it to a set of relation schemas at least in 3NF; specify primary keys and referential integrity constraints for each relation. (5 points)
2. Consider the relation schema $R = (A, B, C, D, E)$ and the set of functional dependencies $F = \{A \rightarrow B, C \rightarrow D, AC \rightarrow E\}$
- (1) List the candidate key(s) for R. Write 'none' if you think there are no candidate keys. (3 points)
 - (2) List the FDs in F that violates BCNF. (3 points)
 - (3) Is R in 3NF? Why? (3 points)

评阅教师	得分

IV. Concurrent Control (10 points total)

Consider two concurrent schedules **S1** and **S2** of transactions T1 and T2.

T1	T2
read (A)	
write (A)	
	read (A)
	write (A)
read (B)	
write (B)	
	read (B)
	write (B)
commit	
	commit

S1

T1	T2
read (A)	
	read (A)
	write (A)
	read (B)
write (A)	
read (B)	
write (B)	
commit	
	write (B)
	commit

S2

1. Is the schedule **S1** conflict serializable? If so, give an equivalent serial schedule. If not, give an explain briefly. (5 points)
2. Is the schedule **S2** conflict serializable? If so, give an equivalent serial schedule. If not, give an explain briefly. (5 points)

评阅教师	得分

V. Database Design (20 points total)

To develop an employee training database system (员工培训数据库系统) for a company. It involves the following situations about employees, instructors, training projects, and training materials.

Every employee has a unique ID, a name, and his/her date of birth (DOB). An instructor has a unique ID, a name and an address. Each training project has a different training number, content, a start date and an end date. A material has a title and description. An employee can take part in several training projects. Each training project has several instructors and some kinds of materials. A material is written by only one instructor and can be used in different projects.

1. Construct an E-R diagram that captures the information above. **(10 points)**
2. Convert the E-R diagram to 3NF relations. Specify keys and referential integrity constraints. **(10 points)**